Hassan Imani

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Machine Learning Engineer with a PhD in Mechanical Engineering and experience in container-based model serving, C++ scientific simulation, and parallel computing with OpenMP and Open MPI. Built production ML pipelines and a hybrid CFD–MD solver; developed molecular simulation methods and graph neural networks for physical systems.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer / Data Scientist — Jul 2025 – Present
Scotiabank | AML AI Research & Special Projects — *Toronto, ON*

- **Training & Inference Optimization:** Refactored conditional generative anomaly detection pipelines for computational efficiency, maintainability, and numerical stability; integrated system-level profiling and performance logging, stabilized training loops, and evaluated loss functions and architecture variants.
- **Experimentation & Correctness:** Built custom hyperparameter optimization tooling, proxy evaluation metrics, and synthetic verification datasets to benchmark unsupervised models despite delayed investigation feedback.
- **Representation Learning:** Built self-supervised sequence models on transaction histories and evaluation suites measuring embedding consistency against customer demographics and risk cohorts.

Machine Learning Engineer — Feb 2024 – Jul 2025
Zown — *Toronto, ON*

- **End-to-End ML Systems:** Owned production ML pipelines; built a two-tower recommendation system with constraint filtering, behavioral feedback, micro-batch training, and automated holdout validation gates before model promotion.
- **Container-Based Model Serving:** Deployed and served property valuation and recommendation models through containers; added continuous training to the property valuation pipeline.
- **Production Inference & Agent Tools:** Integrated a fine-tuned multi-task Vision Transformer into listing ingestion; implemented contract-population and DocuSign tool workflows for a WhatsApp agent.

SCIENTIFIC SIMULATION & MACHINE LEARNING RESEARCH

Doctoral Researcher | Mechanical Engineering — 2021 – 2025
York University — *Toronto, ON*

- **Molecular Dynamics & Analysis:** Developed trajectory-analysis pipelines and spatio-temporal tracking of water-in-oil droplet coalescence, quantifying molecular aggregation, interfacial adsorption, and multiphase dynamics.
- **Simulation Validation:** Developed a regression method for partial molar volumes from molecular dynamics trajectories; validated benchmark simulations against experiments, achieving 2.49% mean relative deviation, reduced to 1.18% with localized sampling.
- **Scientific ML:** Fine-tuned equivariant graph neural networks on small density functional theory (DFT) datasets for adsorption energy and force prediction; evaluated layer freezing and data augmentation through ablation studies.

Master's Researcher | Mechanical Engineering — 2018 – 2021
Tehran Polytechnic — *Tehran, IR*

- **C++ Solver Development:** Built a hybrid computational fluid dynamics–molecular dynamics (CFD–MD) solver using OpenFOAM and ran coupled scientific simulations.
- **Parallel Computing:** Implemented parallel simulation code using OpenMP and Open MPI, including domain decomposition to distribute the computational workload.

EDUCATION

York University — PhD, Mechanical Engineering — 2021 – 2025
Tehran Polytechnic — MSc, Mechanical Engineering — 2018 – 2021
Sharif University of Technology — BSc, Mechanical Engineering — 2013 – 2018

TECHNICAL SKILLS

Systems & Parallel Computing: Docker, container-based model serving, OpenMP, Open MPI, GCP, Spark / PySpark
Scientific Computing & ML: OpenFOAM, CFD–MD coupling, molecular dynamics, PyTorch, PyTorch Geometric
Languages & Tools: C++, Python, SQL, Bash, Git; system profiling, automated validation, agent tool integration

SELECTED PUBLICATIONS

HI Parashkooh, C Jian (all papers).

Graph Neural Networks for Sustainable Energy: Predicting Adsorption in Aromatic Molecules. *ChemE*, 2025.
An Innovative Method for Calculating PMV from a Standard Molecular Trajectory. *J. Mol. Liq.*, 2024.
Data Mining Guided Molecular Investigations on the Coalescence of Water-in-Oil Droplets. *Energy Fuels*, 2022.